CO1 - AT2 Concept Mapping

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1) Student Performance Analysis

1. Data collection
   - Attendance records
   - Internal marks
   - Assignment scores
   - Final examination results
2. Data Preprocessing
   - Remove deplicate records
   - Handle missing values
   - Correct errors in data
3. Data Integration
   - Combine attendance, Marks, and assignment data into one dataset
4. Data transformation
   - Normalize numerical values
   - Convert data into a suitable format
   - Calculate total and Average Marks
5. Data Analysis
   - Analyze attendance percentage
   - Compare student Marks
6. Result
   - Identify weak students
   - Predict academic Performance
   - Suggest Improvements

2) Hospital Patient Data collection

1. Data collection Sources
   - Patient registration forms
   - laboratory reports
   - Medical devices
   - Doctor's notes

2. Data Preprocessing
   - Remove duplicate Patient records
   - Handle missing Information
   - correct data Errors

3. Data Integration
   - Combine data from all sources
   - Create a unified Patient dataset

4. Data Transformation
   - Encode categorical data
   - organize Patient Information

5. Data Analysis
   - Identify diseases
   - Monitor Patient Health
   - Analyze treatment results

6. outcome
   - support diagnosis
   - Improve treatment Planning
   - Help doctors make decisions.

3) Online Shopping Dataset

1. Identify Data Problems →
   - Missing customer details
   - Duplicate orders
   - wrong Product Entries

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2. Data cleaning steps →
   - Handle missing values
   - Remove duplicate records
   - correct Invalid Product Entries

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3. Data validation →
   - check data types
   - verify Order details
   - check consistency and accuracy

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4. clean Dataset →
   - Accurate customer information
   - correct order records
   - consistent Product details

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5. Data Analysis →
   - Study customer Purchasing Patterns
   - Analyze sales
   - Improve business decisions

4)

Smart city Traffic Management

1. Data collection sources →
   - CCTV camera
   - GPS devices
   - Traffic sensors
   - Mobile applications
2. Data Preprocessing →
   - Remove duplicate dates
   - Handle missing values
   - Remove incorrect readings
3. Data Integration →
   - Match timestamps and locations
   - Create a single traffic dataset
4. Data transformation →
   - Convert data into common formats
   - organize location information
5. Data Analysis →
   - Monitor traffic density
   - Identify congestion
   - Analyze traffic patterns
6. outcomes →
   - Optimize traffic signals
   - Improve road safety
   - Reduce congestion

5)

Loan Approval Prediction

1. Data collection → . Income
   . Age
   . Job type
   . Loan amount
   . credit score
2. Data PreProcessing → . Handle missing values
   . Remove duplicate records
   . Correct invalid data
3. Data transformation Techniques → . Encoding: Convert Job type into numerical values
   . Feature scaling: Bring numerical features to comparable scales
4. Feature selection → . Select important features such as
   . Income loan, amount credit history
5. Model Training → . Train a machine learning model
   . using historical loan data
6. Prediction → . Predict loan Approval or rejection
   . support faster and more consistent loan decisions.